Optimizing Incident Management Systems

An Operations Research & Machine Learning Approach

Divagar S (CB.SC.U4AIE23223)
Abhishek Pandey (CB.SC.U4AIE23205)
Prem (CB.SC.U4AIE23241)

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Problem Statement

Core Idea

Predict how long a ticket will take and how satisfied the customer may be, then use Optimization and Queueing Theory to decide who should handle the ticket and how many resolvers should be available.

- **Inefficient Ticket Routing:** Incident routing in support systems is often handled manually or via static rules, leading to processing bottlenecks.
- **Resolver Burnout:** Uneven workload distribution among the 9 active resolvers causes high queue times and resolver fatigue.
- **SLA Violations:** High-priority incidents suffer from delayed assignments, violating Service Level Agreements (SLAs).
- **Low Customer Satisfaction:** The average customer satisfaction is currently **3.23 / 5**, highlighting the need for process optimization.

Objectives

Primary Goal

Optimize the end-to-end incident management workflow to improve system throughput and customer experience by optimizing which resolver should receive this ticket so that overall resolution time is low, workload is balanced, SLA violations are minimized, and customer satisfaction is high.

Operational Efficiency

- Minimize total incident resolution time.
- Minimize SLA violation rate.

Resource & Quality Management

- Balance workloads across resolvers.
- Maximize customer satisfaction scores.

Dataset Overview

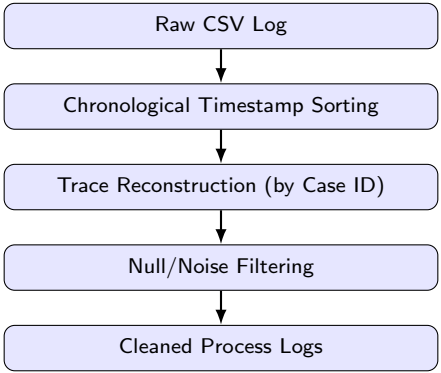
Process Mining Event Log - Incident Management
Work with a realistic event log for incident management.

About the Dataset:

This realistic incident management event log simulates a common IT service process and includes key inefficiencies found in real-world operations. You'll uncover SLA violations, multiple reassignments, bottlenecks, and conformance issues—making it an ideal dataset for hands-on process mining, root cause analysis, and performance optimization exercises.

Metric	Value
Total Events	242,901
Unique Cases	31,588
Resolvers	9
Priorities	High, Med, Low
Avg. CSAT	3.23 / 5.00

Data Preprocessing Flow

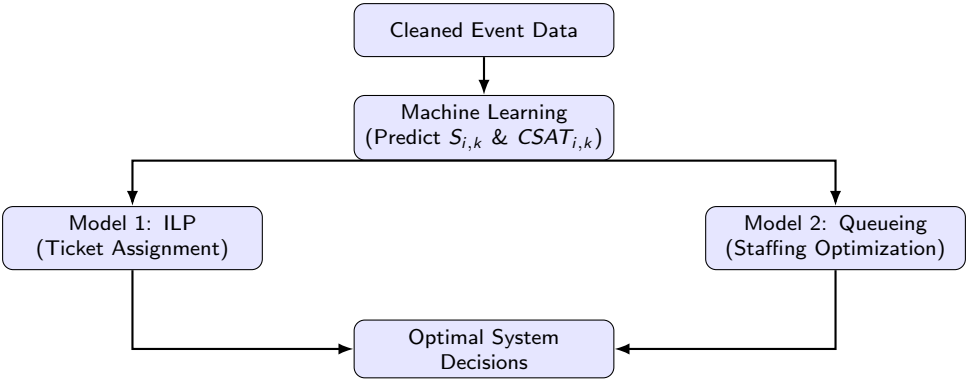


Feature Extraction

To feed both machine learning and optimization models, we extract three categories of features:

- **Temporal Features (Durations):**
  - **Queue Time (W_q):** Time from ticket assignment to WIP (work-in-progress).
  - **Service Time (S):** Time from WIP to ticket solution/escalation.
  - **Total Cycle Time (T):** Full duration from creation to closure.
- **Categorical Context:**
  - Ticket Priority, Issue Type, and Report Channel.
- **Resource Performance Metrics:**
  - Resolver throughput rate ($\mu_{i,j}$) and workload backlog size (W_i).

Methodology



Model 1: Prescriptive Ticket Assignment (ILP)

Optimal Ticket-to-Resolver Matching

- **Purpose:** Assign incoming incidents to the 9 resolvers in real-time, minimizing resolution time and keeping workloads balanced.
- **Optimization Steps:**
  1. Predict service time ($S_{i,k}$) for candidate resolver i and ticket k using regression.
  2. Query current backlog workload (W_i) for all active resolvers.
  3. Construct the ILP coefficient matrix based on predicted service times and priorities.
  4. Run the ILP solver at each assignment decision epoch to compute optimal assignments.

Model 2: Tiered Staffing Optimization

Queuing Network Analysis

- **Purpose:** Model Level 1, 2, and 3 tiers as a network of queues to optimize the number of agents needed to meet target response times.
- **Optimization Steps:**
  1. Estimate Poisson arrival rates (λ) and exponential service rates (μ) for each tier.
  2. Formulate the system as a Jackson Network with routing probabilities ($P_{1\to 2}$ escalation rates).
  3. Solve the multi-server queue equations ($M/M/c$) for steady-state performance.
  4. Optimize the count of active resolvers (c_1, c_2, c_3) to minimize operating costs while keeping average wait time below SLA limits.

Objective Function

We formulate a multi-objective function to balance speed, customer satisfaction, and resolver workloads:

Weighted Objective Function

$$\min \quad Z = \alpha \sum_{k \in K} \sum_{i \in I} P_k \cdot S_{i,k} \cdot x_{i,k} - \beta \sum_{k \in K} \sum_{i \in I} CSAT_{i,k} \cdot x_{i,k} + \gamma \cdot \sigma^2_{\text{workload}}$$

Where:

- $\sum P_k \cdot S_{i,k} \cdot x_{i,k}$: Total priority-weighted resolution time (to minimize).
- $\sum CSAT_{i,k} \cdot x_{i,k}$: Total expected customer satisfaction (to maximize).
- $\sigma^2_{\text{workload}}$: Variance of workloads among resolvers (to minimize for balance).
- α, β, γ: Scaling weights adjusting priorities.

Constraints and Decision Variables

Decision Variable:

$$x_{i,k} = \begin{cases} 1 & \text{if ticket } k \text{ is assigned to resolver } i \\ 0 & \text{otherwise} \end{cases}$$

Constraints:

- **Single Assignment Constraint:** Each incident must be assigned to exactly one resolver.

$$\sum_{i \in I} x_{i,k} = 1 \quad \forall k \in K$$

- **Resolver Capacity Constraint:** No resolver can exceed their maximum workload capacity C_i.

$$W_i + \sum_{k \in K} x_{i,k} \leq C_i \quad \forall i \in I$$

- **Tier Routing Constraint:** Tickets must be assigned only to resolvers active in the correct tier (L1, L2, L3).

Evaluation Metrics

Operations Research Metrics

- **Average Cycle Time:** Mean resolution time of tickets.
- **SLA Compliance Rate:** Percentage of incidents resolved within deadline.
- **Workload Gini Coefficient:** Metric for measuring workload distribution equality.

Machine Learning Metrics

- **Mean Absolute Error (MAE):** On predicted service times.
- **R-squared (R^2):** Variance explained in satisfaction score predictions.

Tools and Technology

- **Programming Language:** Python 3.x
- **Data Manipulation:** Pandas, NumPy (for data processing)
- **Predictive Modeling:** Scikit-Learn, LightGBM (for service time and CSAT predictions)
- **Mathematical Optimization Solvers:** PuLP / Gurobi / SciPy (for ILP optimization)
- **Simulation Engine:** SimPy (for queueing network discrete event simulation)
- **Typesetting & Graphics:** LaTeX/ Beamer (slides), TikZ (flowcharts)

Thank You

Questions & Comments